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Air data probe electronics housing with thermal isolating features

Abstract

In some applications, aircraft air data probes are heated to prevent rain, ice, or other moisture from attaching to the air data probe, ensuring proper functionality of the air data probe. But the elevated temperatures can have negative effects on the electronic components positioned within the air data probe. Therefore, thermal isolating features are added to a housing to thermally isolate the heated parts of the air data probe from the electronic components within the air data probe, which are required to stay relatively cool for proper functioning.

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Background/Summary

BACKGROUND

- (1) The present disclosure relates generally to air data probes on an aircraft, and more particularly to an air data probe electronics housing with thermal isolating features.
- (2) Some aircraft air data probes are heated to prevent rain, ice, or other moisture from attaching to the probe, which could negatively impact the functionality of the air data probe. In many cases, the heat is elevated to extremely hot temperatures to ensure there is no ice accretion on the air data probe during flight of the aircraft. However, there are materials and components within the air data probes in which excess heat is detrimental. As such, there is a need for preventing damage to the materials and components positioned within the heated air data probes.

SUMMARY

- (3) According to one aspect of the disclosure, an air data probe is disclosed. The air data probe includes a body comprising an inlet fluidly coupled to an outlet. A heating element is positioned within the body of the air data probe, and the heating element is configured to increase a temperature of the body. A sensing probe is positioned within the body, and the sensing probe includes a sensing base coupling the sensing probe to the body. A housing is positioned within and coupled to the body. The housing includes a housing body including an upper surface, a lower surface, an inner surface, an outer surface, an inner wall, and an outer wall. The inner wall is offset from the outer wall such that a gap is positioned between the inner wall and the outer wall.
- (4) According to another aspect of the disclosure, a housing for surrounding electrical components within an air data probe configured to gather data during flight of an aircraft is disclosed. The housing includes an upper surface, a lower surface, an inner surface, an outer surface, a rim positioned adjacent the upper surface, and a lower body positioned adjacent the lower surface. An inner wall is offset from an outer wall such that a gap is positioned between the inner wall and the outer wall. The inner wall and the outer wall converge and are coupled adjacent the lower body of the housing body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a side view of an exemplary air data probe used on an aircraft.
- (2) FIG. 1B is a partial cross-sectional view of a housing within the exemplary air data probe taken along Section A-A of FIG. 1A.
- (3) FIG. 2A is a perspective view of an exemplary housing.
- (4) FIG. 2B is a cross-sectional view of the exemplary housing taken along Section B-B of FIG. 2A.

DETAILED DESCRIPTION

- (5) FIG. 1A is a side view of air data probe **10** used on an aircraft (not shown). FIG. 1B is a partial cross-sectional view of housing **12** positioned within air data probe **10** taken along Section A-A of FIG. 1A. FIGS. 1A-1B will be discussed together. Further, hereinafter, air data probe **10** will be referred to as probe **10**, but it is to be understood that probe **10** is referring to air data probe **10**. Probe **10** is a device which detects or measures a physical property and records, initiates, or otherwise responds to the detected or measured physical property. In some examples, probe **10** can be used to detect the total air temperature, air pressure, air velocity, and/or air density, among other properties not specifically described, during flight of an aircraft. Further, in some examples, probe **10** can be positioned at the nose cone of an aircraft such that probe **10** is positioned partially within the skin of the aircraft and partially outside the skin of the aircraft.
- (6) Since a portion of probe **10** is positioned outside the skin of the aircraft, probe **10** can experience harsh operating conditions including, but not limited to low operating temperatures, ice, rain, sleet, snow, etc. In some examples, to ensure proper functionality of probe **10**, probe **10** can be

heated to prevent ice, rain, or other precipitation from accumulating on probe **10** during flight of the aircraft. Heating probe **10** can melt ice on probe **10**, allowing the precipitation to blow off probe **10** during flight conditions. Further, heating probe **10** can facilitate evaporation of any precipitation on probe **10** during flight conditions. Probe **10** is configured to gather air data during flight of the aircraft, providing the pilots of the aircraft critical information during flight conditions. As such, it is important that probe **10** is functioning properly and providing the pilots of the aircraft accurate information.

(7) Referring to FIG. **1A**, probe **10** includes body **14**, flange **16**, electrical connector **18**, inlet **20**, and outlet **22**. Body **14** is the main structure of probe **10** that protects the components within probe **10** from environmental conditions. In some examples, body **14** can be constructed from at least one metallic material. In other examples, body **14** can be constructed from a ceramic material, a polymeric material, and/or a composite material. With that said, the following discussion with focus on an embodiment in which body **14** is constructed from a metallic material. Body **14** includes inlet **20** positioned at a distal end of probe **10** and outlet **22** positioned generally at a central portion of probe **10**. Inlet **20** is positioned at a first distal end of probe **10**. More specifically, inlet **20** is positioned at the distal end of probe **10** positioned outside the skin of the aircraft. Inlet **20** can be an aperture extending through body **14**, allowing air from outside the aircraft to enter and flow through inlet **20** into body **14**. The air that flows through inlet **20** proceeds by flowing through body **14** and then exits body **14** by flowing outwards through outlet **22**. Outlet **22** can be an aperture that extends through body **14**, allowing air from within body **14** to flow out through outlet **22** to exit probe **10**. In some examples, as shown, outlet **22** can be positioned generally at the central portion of probe **10** adjacent flange **16**.

(8) Flange **16** is positioned adjacent outlet **22** generally at a central portion of probe **10**. Flange **16** extends outwards from the main portions of body **14** and flange **16** extends fully around an outer surface of body **14** of probe **10**. As such, flange **16** is a flat rim or collar extending outwards from body **14** around an exterior of body **14**. In some examples, flange **16** can have a circular cross-section viewing in the axial direction along probe **10**. In other examples, flange **16** can have a cross-section of any geometric shape. Flange **16** is the feature of probe **10** that separates the portions of probe **10** positioned within and positioned outside the skin of an aircraft. More specifically, the portion of probe **10** including inlet **20** and outlet **22** is positioned on the outside of the skin of the aircraft and the portion of probe **10** including electrical connector **18** is positioned on the inside of the skin of the aircraft. Electrical connector **18** can be any electrical connector that creates an electrical connection between probe **10** and electronics on an aircraft. As shown in FIG. **1A**, in some examples electrical connector **18** can include a threaded connection for securing probe **10** to an aircraft electrical system.

(9) Referring to FIG. **1B**, which is a partial cross-sectional view of probe **10** taken along Section A-A of FIG. **1A**, probe **10** also includes heating element **24**, sensing probe **26**, sensing base **28**, electrical wires **30**, electrical leads **32**, housing **12**, and potting **34**, each of which are positioned within body **14** of housing **12**. In the example shown, probe **10** includes one heating element **24** wrapped around an outer surface of probe **10**. In another example, probe **10** can include more than one heating element **24**. In the example shown, heating element **24** is wrapped around an outer surface of probe **10** to facilitate melting of ice or evaporation of fluid on outer surfaces of probe **10**. Heating element **24** can be an electrical resistance heating element that increases in temperature when an electrical current is transferred to the electrical resistance heating element. In some examples, heating element **24** can be electrical wires or electrical pads, among other options, that increase in temperature when an electrical current is transferred to heating element **24**, such that heating element **24** increases the temperature of body **14**.

(10) Sensing probe **26** and sensing base **28** are components of probe **10** positioned within body **14** and generally aligned with a central axis of probe **10**. Sensing probe **26** is the component of probe **10** which detects or measures a physical property of the air flowing through probe **10**. More

specifically, air from outside the aircraft flows through inlet **20** into body **14** and follows the flow path, indicated by the arrows in FIG. **1B**, through body **14** to outlet **22**. As such, the air flowing through probe **10** flows past sensing probe **26** within body **14** and sensing probe **26** can gather the desired data from the air flow before the air exits probe **10** through outlet **22**. Sensing probe **26** is coupled to sensing base **28** to secure sensing probe **26** in a fixed position during operation of probe **10**. Further, sensing base **28** is coupled to internal surfaces or features of body **14** to secure both sensing base **28** and sensing probe **26** in a fixed position during operation of probe **10** on an aircraft.

(11) Electrical wires **30** are electrically coupled to and extend from a distal end of sensing probe **26** positioned within housing **12** of probe **10**. Electrical wires **30** are configured to transfer data gathered by sensing probe **26** to an electrical system and/or a controller on an aircraft. More specifically, a plurality of electrical wires **30** are electrically coupled to a plurality of electrical leads **32**, and the plurality of electrical leads **32** are coupled to electrical connector **18** of probe **10**. As such, air data gathered by sensing probe **26** is transferred through electrical wires **30**, through electrical leads **32**, through electrical connector **18**, and to an electrical controller system of an aircraft. Therefore, electrical wires **30**, electrical leads **32**, and electrical connector **18** allow the transfer of data from sensing probe **26** to an aircraft, and breaking the electrical connection between any of the components can prevent the transfer of data from sensing probe **26** to an aircraft. As such, maintaining the electrical connection between electrical wires **30**, electrical leads **32**, and electrical connector **18** is critical to maintaining proper functioning of probe **10**.

(12) Housing **12** is positioned within body **14** of probe **10** and housing **12** is coupled to body **14** of probe **10**. In some examples, housing **12** can be coupled to internal surfaces or features of body **14** through a welded connection. As such, in some examples, housing **12** can be constructed from a metallic material. In other examples, housing **12** can be constructed from a non-metallic material and housing **12** can be coupled to body **14** through an adhesive, a mechanical feature, or a fastener, among other options. Further, in some examples, housing **12** can be axially aligned with a central axis of probe **10**. Therefore, in some examples, housing **12** can be positioned concentric with body **14** of probe **10**. Housing **12** is fixedly positioned within body **14** of probe and adjacent sensing base **28** of probe **10**. Further, housing **12** is positioned such that housing **12** surrounds the plurality of electrical wires **30** extending from sensing probe **26**. More specifically, housing **12** is positioned such that housing **12** surrounds a connection point between the plurality of electrical wires **30** and the plurality of electrical leads **32** such that the plurality of electrical wires **30** and the plurality of electrical leads **32** extend within a center opening within housing **12**. In addition, the connection point between the plurality of electrical wires **30** and the plurality of electrical leads **32** is encompassed by potting **34**.

(13) Potting **34** is positioned within an inner surface of housing **12** and potting **34** is positioned adjacent an upper surface of housing **12**. In some examples, as shown in FIG. **1B**, potting **34** extends only a partial distance from an upper surface of housing **12** towards a lower surface of housing **12**. In other examples, potting **34** can extend a full distance from an upper surface of housing **12** to a lower surface of housing **12**. Further, potting **34** can encompass and fully surround the connection point between the plurality of electrical wires **30** and the plurality of electrical leads **32**. As such, potting **34** is configured to aid in securing the connection between the plurality of electrical wires **30** and the plurality of electrical leads **32**. In some examples, potting **34** can provide vibration support to the plurality of electrical wires **30** and the plurality of electrical leads **32**, and potting **34** can act as an electrical isolator between the plurality of electrical wires **30**, the plurality of electrical leads **32**, and housing **12**. In some examples, potting **34** can be constructed from a high-temperature epoxy material, such that potting **34** is introduced into housing **12** as a fluid and then solidifies to produce the vibration support and electrical isolation.

(14) Although potting **34** can be constructed from a high-temperature epoxy material, the heat produced by heating element **24** of probe **10** can have undesirable effects on potting **34**. For

example, some high-temperature epoxy materials are designed to prevent melting under high temperatures, but if a high enough temperature is reached the high-temperature epoxy material could crack due to its brittle material properties. Further, potting 34 is likely to have different material properties than the metallic components surrounding potting 34, such as housing 12 and body 14 of probe 10. Therefore, potting 34 is likely to have a different coefficient of thermal expansion than housing 12 and when heat is applied to housing 12 and potting 34, the components will expand at differing rates. High-temperature epoxy material generally expand at a greater rate than metallic materials due when heat is applied to the components. As such, when heat is applied to potting 34 and housing 12, potting 34 can expand at a greater rate than housing 12 which can cause potting 34 to expand outside its intended adhesion area. In some scenarios, potting 34 can expand to an extent that it pulls apart and breaks the connection between the plurality of electrical wires 30 and the plurality of electrical leads 32. In turn, this can sever the electrical connection between probe 10 and an aircraft, preventing data from transferring from probe 10 to the aircraft. To remedy this potential issue, housing 12 includes features preventing damage to the plurality of electrical wires 30 and the plurality of electrical leads 32.

(15) FIG. 2A is a perspective view of housing 12 shown in FIG. 1B. FIG. 2B is a cross-sectional view of housing 12 taken along Section B-B of FIG. 2A. FIGS. 2A-2B will be discussed together. Housing 12 includes housing body 36 including upper surface 38, lower surface 40, inner surface 42, outer surface 44, rim 46, lower body 48, first alignment feature 50, second alignment feature 52, inner wall 54, and outer wall 56. Upper surface 38 is a surface at a first distal end of housing 12 and lower surface 40 is a surface at a second distal end of housing 12, opposite first distal end. As shown best in FIG. 2B, inner surface 42 is the surface positioned within the walls constituting housing 12 and outer surface 44 is the surface positioned outside the walls constituting housing 12. In the example shown in FIGS. 2A-2B, housing 12 includes a circular cross-section in the axial direction such that housing 12 has a generally cylindrical shape. In other examples, housing 12 can have an axial cross-section of any geometrical shape.

(16) Rim 46 is positioned adjacent upper surface 38 of housing 12 and rim 46 extends fully around a perimeter or circumference of housing 12, about central axis CL of housing 12. Rim 46 is a ledge or cutout that extends inwards from outer surface 44 towards central axis CL of housing 12. As such, an inner surface or edge of rim 46 has a smaller diameter than outer surface 44 of housing 12. Rim 46 is configured to aid in the proper positioning of housing 12 within body 14 of probe 10. Lower body 48 is positioned adjacent lower surface 40 of housing 12 and lower body 48 extends fully around outer surface 44 of housing 12, about central axis CL of housing 12. Lower body 48 is a portion of housing 12 that extends outwards from outer surface 44 of housing 12 away from central axis CL of housing 12. As such, lower body 48 has a larger diameter than outer surface 44 of housing 12. Further, in the example shown, an outer diameter of rim 46 is less than an outer diameter of lower body 48. Lower body 48 having a larger diameter than rim 46 and outer surface 44 aids in housing 12 being correctly positioned within body 14 of probe 10.

(17) As shown in FIGS. 2A-2B, housing body 36 also includes first alignment feature and second alignment feature 52. First alignment feature 50 is positioned adjacent rim 46 of housing 12 and second alignment feature 52 is positioned adjacent lower body 48 of housing 12. First alignment feature 50 is a cutout or indent extending into housing body 36 adjacent rim 46 and upper surface 38 of housing 12. Second alignment feature 52 is a protrusion extending outwards from outer surface 44 of housing body 36. In the example shown, first alignment feature has a generally rectangular shape, but in another example first alignment feature 50 could have any geometric shape. Further, in the example shown, second alignment feature 52 is a generally triangular shaped protrusion, but in another example second alignment feature 52 could be any geometric shaped protrusion. First alignment feature 50 and second alignment feature 52 are vertically aligned with respect to a vertical plane extending from lower surface 40 of housing 12 to upper surface 38 of housing 12. First alignment feature 50 and second alignment feature 52 aid in housing 12 being

correctly positioned within body **14** of probe **10**. More specifically, first alignment feature **50** and second alignment feature **52** are configured to mate with mating features of body **14** of probe **10** to ensure housing **12** is oriented in the correct position while being inserted into body **14** of probe **10**. (18) As shown best in FIG. 2B, housing **12** includes inner wall **54** and outer wall **56**. Inner wall **54** is generally a hollow cylinder extending from lower body **48** to upper surface **38** of housing **12**. More specifically, inner wall **54** extends a partial distance from lower body **48** and outer wall **56** of housing **12** towards central axis CL and then inner wall **54** angles upward such that a portion of inner wall **54** is parallel with central axis CL. In the example shown, the portion of inner wall **54** parallel with central axis CL is a hollow cylinder with a circular cross-section. In another example, inner wall **54** can have a cross-section of any geometric shape. Outer wall **56** is generally a hollow cylinder that extends from lower body **48** to upper surface **38**. Outer wall **56** is oriented parallel to central axis CL and outer wall **56** is concentric with inner wall **54**. In the example shown, outer wall **56** is a hollow cylinder with a circular cross-section. In another example, outer wall **56** can have a cross-section of any geometric shape.

(19) Outer wall **56** fully surrounds an outer circumference of inner wall **54**. More specifically, inner wall **54** is offset from outer wall **56** in a direction towards central axis CL such that a gap is positioned between inner wall **54** and outer wall **56**. Further, a portion of inner wall **54** converges with outer wall **56** such that inner wall **54** and outer wall **56** are coupled adjacent lower body **48** of housing **12**. The gap between inner wall **54** and outer wall **56** is an air gap in which air or other gas is free to flow within, discussed further below. In the example shown, inner wall **54** and outer wall **56** have equal thicknesses. In another example, inner wall **54** and outer wall **56** can have differing thicknesses. The thickness of inner wall **54** and outer wall **56** can differ depending on requisite physical properties of housing **12**.

(20) Central aperture **58** is formed by inner surface **42** of inner wall **54**. Central aperture **58** is axially aligned with central axis CL and central aperture **58** extends through housing **12** from upper surface **38** to lower surface **40** of housing **12**. Referring again to FIG. 1B, central aperture **58** is configured to allow a distal end of sensing probe **26**, electrical wires **30**, and electrical leads **32** to be positioned within and extend at least partially through central aperture **58**. In the example shown, central aperture **58** has a circular cross-section in the axial direction of central axis CL. In other examples, central aperture **58** could have any geometric shape as its cross-section in the axial direction of central axis CL. Housing **12** also includes a plurality of apertures **60** extending through upper surface **38** of housing body **36** to the gap positioned between inner wall **54** and outer wall **56**. The plurality of apertures **60** are configured to allow air or other gas to flow into and out of the gap positioned between inner wall **54** and outer wall **56**.

(21) Housing **12** with inner wall **54** and outer wall **56** can be manufactured using an additive manufacturing process, such as a laser powder bed fusion process. As such, housing **12** is grown in a vertical direction starting at lower surface **40** and finishing at upper surface **38**. In some examples, housing **12** can be constructed from a metallic material. In other examples, housing **12** can be constructed from a ceramic material, a polymeric material, and/or a composite material. The plurality of apertures **60** positioned adjacent upper surface **38** of housing **12** also aid in removing any excess material from the gap between inner wall **54** and outer wall **56** that may have accumulated within the gap during the additive manufacturing process.

(22) Referring again to FIG. 1B, during assembly of probe **10**, housing **12** is pressed into body **14**, housing **12** is then welded or otherwise coupled to body **14**, and electrical wires **30** are soldered or otherwise electrically coupled to electrical leads **32**. Then a liquefied potting **34** is dispensed within inner surface **42** of inner wall **54** of housing **12** and the potting **34** is subsequently cured and allowed to harden within inner surface **42** of inner wall **54** of housing **12**. The cured potting **34** positioned within inner surface **42** of inner wall **54** encompasses and fully surrounds the connection point between electrical wires **30** and electrical leads **32**.

(23) During use of probe **10** on an aircraft, probe **10** is heated through heating element **24** and

potting 34 will increase in temperature and expand in volume. But the gap positioned between inner wall 54 and outer wall 56 acts as a thermal isolator and the air within the gap absorbs some of the heat before the heat can reach potting 34. As such, the gap (thermal isolator) minimizes the amount of heat absorbed by potting 34 and therefore minimizes the amount of expansion of potting 34. Further, potting 34 being positioned only within inner surface 42 of inner wall 54 results in potting 34 having less total volume, as compared to a housing without the inner wall, the outer wall, and the air gap between the walls. In turn, the reduced volume of potting 34 within housing 12 further minimizes the expansion of potting 34 within housing 12, as compared to a potting within a housing without the inner wall, outer wall, and air gap.

(24) As such, including inner wall 54, outer wall 56, and the air gap between the walls within housing 12 hinders potting 34 from growing outside its intended adhesion area. In turn, this prevents potting 34 from expanding too much and breaking apart the electrical connection between electrical wires 30 and electrical leads 32, ensuring the electronic components within probe 10 remain functional during heating of probe 10. Protecting the electronic components within probe 10 makes probe 10 more robust and allows probe 10 to continue working accurately during heating conditions. Probe 10 including inner wall 54, outer wall 56, and the air gap between the walls prevents damage to the electronic components within probe 10 by ensuring the heat induced on probe 10 does not have detrimental effects to the electronic components. As such, probe 10 can be heated to prevent rain and ice buildup and probe 10 will continue transferring accurate data to the pilot and the co-pilot during flight of the aircraft.

Discussion of Possible Embodiments

(25) The following are non-exclusive descriptions of possible embodiments of the present invention.

(26) An air data probe comprising: a body comprising an inlet fluidly coupled to an outlet; a heating element positioned within the body of the air data probe, wherein the heating element is configured to increase a temperature of the body; a sensing probe positioned within the body, wherein the sensing probe includes a sensing base coupling the sensing probe to the body; and a housing positioned within and coupled to the body, wherein the housing includes: a housing body including an upper surface, a lower surface, an inner surface, an outer surface, an inner wall, and an outer wall; wherein the inner wall is offset from the outer wall such that a gap is positioned between the inner wall and the outer wall.

(27) The air data probe of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(28) The housing surrounds a plurality of electrical wires extending from the sensing probe, and wherein the plurality of electrical wires are electrically coupled to a plurality of electrical leads.

(29) A connection point between the plurality of electrical wires and the plurality of electrical leads is encompassed by a potting, and wherein the potting is positioned within the inner wall of the housing.

(30) The potting extends a partial distance from the upper surface of the housing towards the lower surface of the housing, and wherein the potting is constructed from a high-temperature epoxy material.

(31) A plurality of apertures extend through the upper surface of the housing body to the gap positioned between the inner wall and the outer wall of the housing body.

(32) The inner wall and the outer wall of the housing body have equal thicknesses; and the outer wall of the housing body fully surrounds a circumference of the inner wall of the housing body.

(33) The inner wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the outer wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the inner wall and the outer wall are concentric.

(34) A flange extending around an outer surface of the body and an electrical connector positioned

at a distal end of the air data probe opposite the inlet of the air data probe.

(35) The housing is coupled to the body through a welded connection.

(36) The air data probe is a total air temperature probe configured to measure air temperature outside an outer skin of an aircraft.

(37) The following are further non-exclusive descriptions of possible embodiments of the present invention.

(38) A housing for surrounding electrical components within an air data probe configured to gather data during flight of an aircraft, the housing comprising: a housing body including an upper surface, a lower surface, an inner surface, an outer surface, a rim positioned adjacent the upper surface, and a lower body positioned adjacent the lower surface; and an inner wall offset from an outer wall such that a gap is positioned between the inner wall and the outer wall, wherein the inner wall and the outer wall converge and are coupled adjacent the lower body of the housing body.

(39) The housing of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(40) A first alignment feature and a second alignment feature, wherein the first alignment feature is positioned adjacent the rim of the housing body, and wherein the second alignment feature is positioned adjacent the lower body of the housing body.

(41) The first alignment feature and the second alignment feature are vertically aligned with respect to a vertical plane extending from the lower surface to the upper surface of the housing body; the first alignment feature is an indentation extending into the housing body; and the second alignment feature is a protrusion extending outwards from the outer surface of the housing body.

(42) A plurality of apertures extend through the upper surface of the housing body to the gap positioned between the inner wall and the outer wall.

(43) The inner wall and the outer wall have equal thicknesses.

(44) The outer wall fully surrounds a circumference of the inner wall.

(45) The inner wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the outer wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the inner wall and the outer wall are concentric.

(46) The inner wall extends a partial distance from the outer wall of the housing towards a central axis of the housing, such that an aperture aligned with the central axis extends through the housing from the upper surface to the lower surface of the housing.

(47) The housing includes a circular cross-section, such that the housing is generally cylindrical in shape.

(48) An outer diameter of the rim is less than an outer diameter of the lower body.

(49) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. An air data probe comprising: a body comprising an inlet fluidly coupled to an outlet; a heating element positioned within the body of the air data probe, wherein the heating element is configured to increase a temperature of the body; a sensing probe positioned within the body, wherein the sensing probe includes a sensing base coupling the sensing probe to the body; and a housing positioned within and coupled to the body, wherein the housing includes: a housing body including

an upper surface, a lower surface, an inner surface of an inner wall, and an outer surface of an outer wall; wherein the inner wall is offset from the outer wall such that a gap is positioned between the inner wall and the outer wall; wherein the housing surrounds a plurality of electrical wires extending from the sensing probe, and wherein the plurality of electrical wires are electrically coupled to a plurality of electrical leads; wherein a connection point between the plurality of electrical wires and the plurality of electrical leads is encompassed by a potting, and wherein the potting is positioned within the inner surface of the inner wall of the housing body, whereby the potting is hindered from growing outside an intended adhesion area.

2. The air data probe of claim 1, wherein the potting extends a partial distance from the upper surface of the housing towards the lower surface of the housing, and wherein the potting is constructed from a high-temperature epoxy material.

3. The air data probe of claim 1, wherein a plurality of apertures extend through the upper surface of the housing body to the gap positioned between the inner wall and the outer wall of the housing body.

4. The air data probe of claim 1, wherein: the inner wall and the outer wall of the housing body have equal thicknesses; and the outer wall of the housing body fully surrounds a circumference of the inner wall of the housing body.

5. The air data probe of claim 1, wherein the inner wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the outer wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the inner wall and the outer wall are concentric.

6. The air data probe of claim 1 and further comprising a flange extending around an outer surface of the body and an electrical connector positioned at a distal end of the air data probe opposite the inlet of the air data probe.

7. The air data probe of claim 1, wherein the housing is coupled to the body through a welded connection.

8. The air data probe of claim 1, wherein the air data probe is a total air temperature probe configured to measure air temperature outside an outer skin of an aircraft.

9. A housing for surrounding electrical components within an air data probe configured to gather data during flight of an aircraft, the housing comprising: a housing body including an upper surface, a lower surface, an inner surface, an outer surface, a rim positioned adjacent the upper surface, and a lower body positioned adjacent the lower surface; an inner wall offset from an outer wall such that a gap is positioned between the inner wall and the outer wall, wherein the inner wall and the outer wall converge and are coupled adjacent the lower body of the housing body, and a first alignment feature and a second alignment feature, wherein the first alignment feature is positioned adjacent the rim of the housing body, and wherein the second alignment feature is positioned adjacent the lower body of the housing body.

10. The housing of claim 9, wherein: the first alignment feature and the second alignment feature are vertically aligned with respect to a vertical plane extending from the lower surface to the upper surface of the housing body; the first alignment feature is an indentation extending into the housing body; and the second alignment feature is a protrusion extending outwards from the outer surface of the housing body.

11. The housing of claim 9, wherein a plurality of apertures extend through the upper surface of the housing body to the gap positioned between the inner wall and the outer wall.

12. The housing of claim 9, wherein the inner wall and the outer wall have equal thicknesses.

13. The housing of claim 9, wherein the outer wall fully surrounds a circumference of the inner wall.

14. The housing of claim 9, wherein the inner wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the outer wall is a hollow cylinder extending from the lower body to the upper surface, and wherein the inner wall and the outer wall are concentric.

15. The housing of claim 9, wherein the inner wall extends a partial distance from the outer wall of

the housing towards a central axis of the housing, such that an aperture aligned with the central axis extends through the housing from the upper surface to the lower surface of the housing.

16. The housing of claim 9, wherein the housing includes a circular cross-section, such that the housing is generally cylindrical in shape.

17. The housing of claim 9, wherein an outer diameter of the rim is less than an outer diameter of the lower body.
